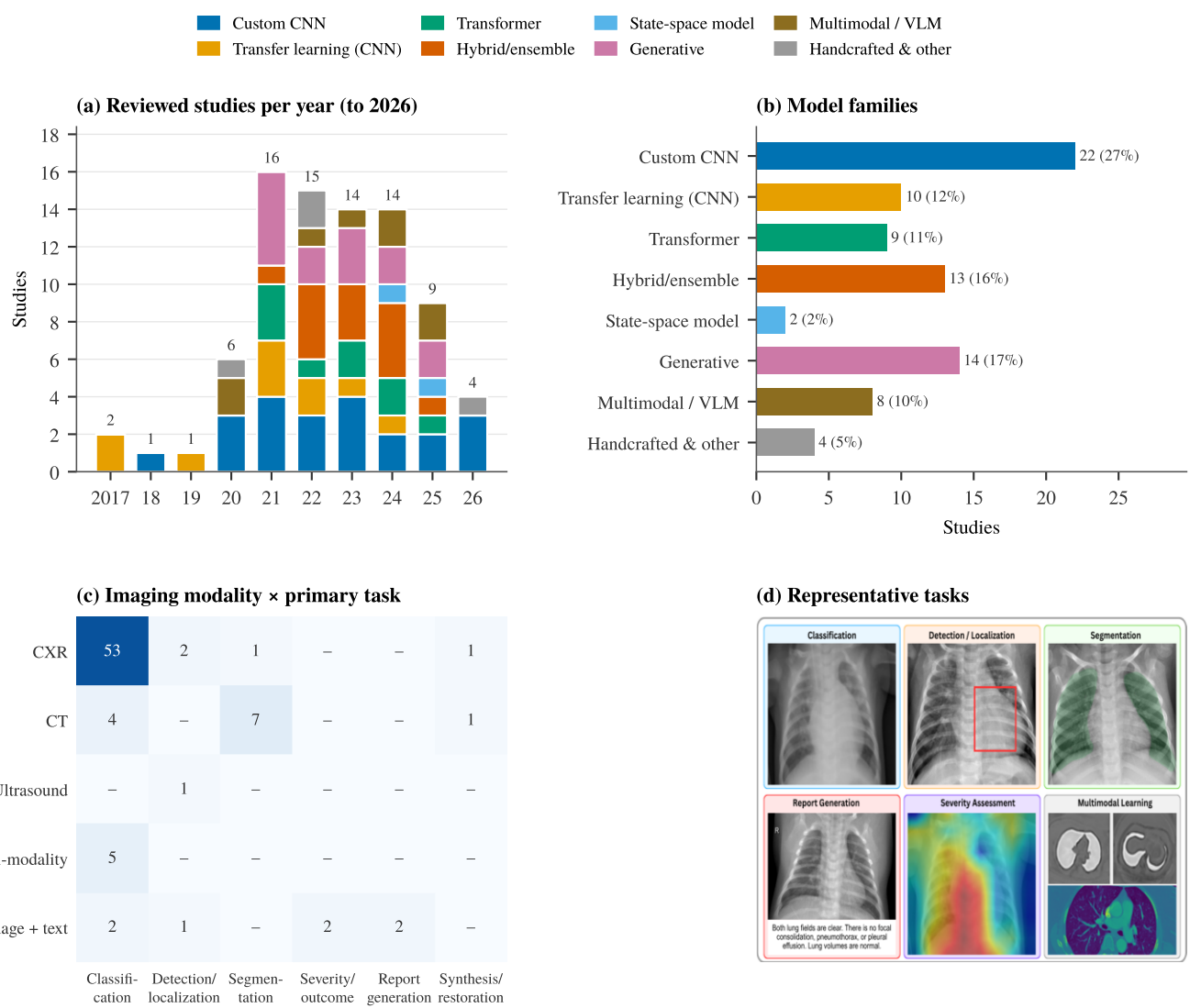

Custom CNN
Transfer learning (CNN)
Transformer
Hybrid/ensemble
State-space model
Generative
Multimodal / VLM
Handcrafted & other
(a) Reviewed studies per year (to 2026)
Studies
2017 18 19 20 21 22 23 24 25 26
2 1 1 6 16 15 14 14 9 4
(b) Model families
Custom CNN 22 (27%)
Transfer learning (CNN) 10 (12%)
Transformer 9 (11%)
Hybrid/ensemble 13 (16%)
State-space model 2 (2%)
Generative 14 (17%)
Multimodal / VLM 8 (10%)
Handcrafted & other 4 (5%)
Studies
(c) Imaging modality × primary task
CXR 53 2 1 – – 1
CT 4 – 7 – – 1
Ultrasound – 1 – – – –
Multi-modality 5 – – – – –
Image + text 2 1 – 2 2 –
Classifi-cation
Detection/ localization
Segmen-tation
Severity/ outcome
Report generation
Synthesis/ restoration
(d) Representative tasks
Classification
Detection / Localization
Segmentation
Report Generation
Severity Assessment
Multimodal Learning
Both lung fields are clear. There is no focal consolidation, pneumothorax, or pleural effusion. Lung volumes are normal.